

ELECTRONICS RETAIL · DATA ANALYTICS CASE STUDY

Sales and Revenue Trends

2016 – 2020

An analysis of sales performance, store performance, and customer trends across the Global Electronics Retailer network.

Yisel Alvarado Ayala

Portfolio Project · 2026

About the Company & Analysis Goals



About the Company

Global Electronics Retailer is a fictional global company specializing in electronic devices, home appliances, and technology solutions. It operates both online and through physical stores worldwide.

50+

Stores
worldwide

2016–2020

Years analyzed

3

Analysis goals

Analysis Goals



Sales Performance

Identify and understand sales performance across product categories.



Store Performance

Evaluate store performance across the global retail network.



Customer Trends

Identify customer and sales trends over time from 2016 to 2020.

Process

A structured, cloud-based workflow was used to collect, store, query, and visualize the data.

01



Download Dataset

Downloaded the Global Electronics Retailer dataset from Maven Analytics.

02



Azure SQL Setup

Created the Azure SQL Database using the Azure for Students free tier and built all tables using the Query Editor.

03



Load Data

Loaded CSV data using a custom Python script (load_data.py) with the pyodbc library, then exported the database as a .bacpac backup.

04



Connect & Query

Connected Azure SQL to VS Code using the mssql and DBCode extensions, then wrote 12 T-SQL queries across three analysis goals.

05



Visualize & Collaborate

Built visualizations using DBCode's charting feature and collaborated via GitHub with a shared repository.

Sales Performance



8

“Silent killer” products identified — high volume but ranked outside the top 20 on profit.

Pricing on these SKUs is the fastest lever for margin recovery.



Home Appliances lead profitability

Home Appliances lead average profit per unit despite not being the highest-volume category.



Computers & Cell Phones: high volume, mid-tier margin

These categories sell the most units but rank mid-tier on profitability — volume isn't translating into proportional profit.

Store Performance

18

High Performers

7

Medium Performers

32

Low Performers



Revenue concentrated in North America

Top revenue stores are concentrated in Canada and the United States, while European stores consistently underperform by comparison.

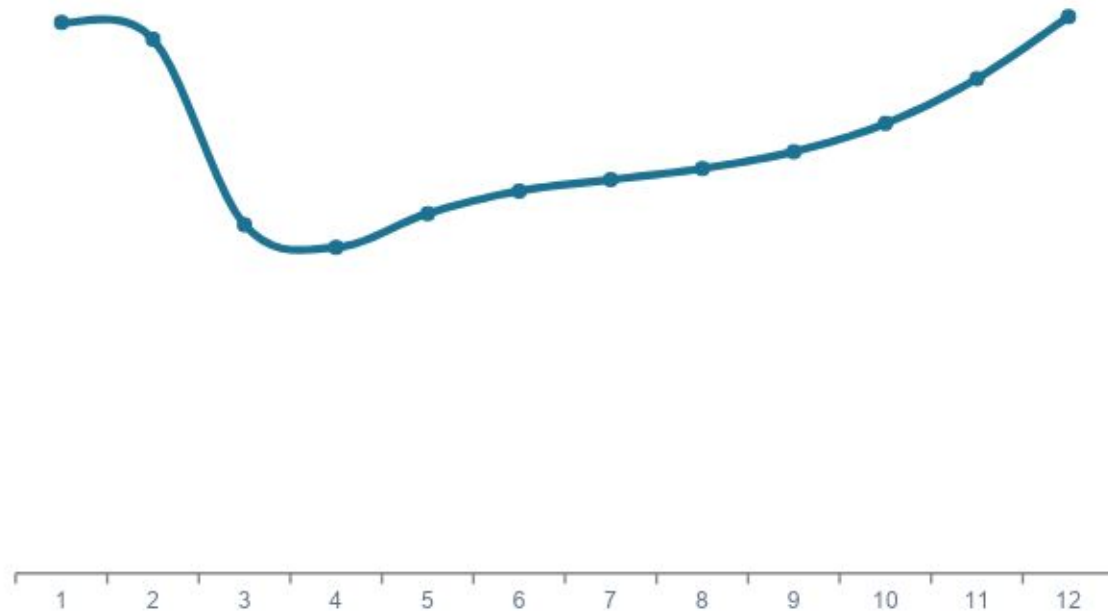


Size doesn't predict revenue

Store size does not strongly correlate with revenue — Wyoming has only 840 sq m but ranks #14 in total revenue network-wide.

Trends Over Time

Illustrative Monthly Sales Pattern



Seasonal peaks and dips

Sales peak in January and February, drop sharply in March–April, then stabilize with a December uptick.



Concentrated customer base

The customer base is concentrated in a handful of countries rather than spread evenly worldwide.



Year-over-year growth

Year-over-year unit growth shows the business is expanding overall despite seasonal volatility.

CONCLUSION & RECOMMENDATIONS

The company is profitable and growing - but there are clear optimization opportunities that can accelerate performance further.

01

PHASE 1

Reprice silent-killer products

Reevaluate pricing on silent-killer products — small margin gains compound fast at high volume.

02

PHASE 2

Replicate top-store practices

Study what makes top-performing stores successful and apply those lessons to Low Performer locations.

03

PHASE 3

Align spend with seasonality

Align marketing budget with seasonal peaks — concentrate promotions around January/February and December.

04

PHASE 4

Expand in underserved markets

Expand in underserved, high-population countries where customer count is low relative to market size.

Data, Tools & Acknowledgements



Dataset

Maven Analytics — Global Electronics Retailer

mavenanalytics.io/data-playground/global-electronics-retailer



GitHub Repository

Shared project repository for collaboration, version control, and code sharing.

github.com/yvalvaradoa-debug



AI Tools & Assistance

Google Gemini

Assistance with documentation.

Claude AI

Assistance with Python script and query structure.

Google Slides

Original presentation tool used for this slide deck.

Thank You

Questions are welcome.



Contact

yyalvaradoa@gmail.com

Yisel Alvarado Ayala